Task 7: Deployment Process Documentation

FastAPI on AWS EC2 | Docker, Nginx, GitHub Actions, GHCR

1. Build and Publish Docker Image

The application is packaged using Docker and published to GitHub Container Registry (GHCR). GitHub Actions automates image building, validation, and publishing when changes are pushed to the `main` branch.

2. Deploy Updated Application

Connect to the AWS EC2 instance using AWS Systems Manager Session Manager and navigate to the project directory.

```
cd /home/ubuntu/fastapi-devops
sudo docker compose pull api
sudo docker compose up -d api
```

3. Verify Deployment

Check the container status:

```
sudo docker compose ps
```

Verify the API endpoints:

- **Welcome:** `https://devops-fastapi.duckdns.org/`
- **Health:** `https://devops-fastapi.duckdns.org/health`
- **Application information:** `https://devops-fastapi.duckdns.org/api/info`

Confirm the application version matches the expected release.

4. Inspect Logs

View application and Nginx logs:

```
sudo docker compose logs --tail=50 api
sudo docker compose logs --tail=50 nginx
```

5. Troubleshooting

If deployment fails, inspect the container status and logs. Check image availability, GitHub Actions results, Nginx configuration, TLS certificates, and security-group rules.

6. Deployment Recovery

Track previously deployed images using unique version tags or image digests. If a deployment fails, restore the last known-good image and verify the application health endpoint.

A controlled test should demonstrate a failed health check followed by successful recovery. This recovery procedure must be implemented and tested before it can be considered complete.